

Jeroen Van Goey

Staff Research Engineer in BioAI

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Summary

Machine learning research engineer working at the intersection of AI and science, bridging scientific ML, biological domain knowledge, production-quality software, and team leadership. I build generative models and take them from prototype to deployment, with a focus on *de novo* peptide sequencing.

Author of 8 publications, including in *Nature Machine Intelligence*.

Currently looking for Staff/Principal Research Engineer roles in AI for science. Based in Cape Town, open to remote-first roles now and hybrid/on-site roles in Europe from 2027.

Experience

InstaDeep, Staff Research Engineer in BioAI

Cape Town, South Africa

Aug 2022 – present

3 years 11 months

- **Tech lead and hiring manager** for two cross-disciplinary **ML research teams** of about 5 to 8 people each: *de novo* peptide sequencing (with the Technical University of Denmark) and signal-peptide design for secretion efficiency (with BioNTech).
- Delivered a **client collaboration with Syngenta**, applying **genomic language models** (AgroNT, a Nucleotide Transformer trained on ~10.5 million genomic sequences spanning trillions of base pairs across 48 plant species) to accelerate crop trait research.
- Set **research direction** and take models **from research to production**, balancing scientific rigour with reliable, maintainable engineering.
- **Lead the BioAI department** of InstaDeep's Cape Town office and serve as the office's **site manager** (an office of about 25 people); hire and grow the BioAI teams across the Cape Town and Kigali offices.
- Built the team's **ML engineering foundations**: scalable, Transformer-based libraries (Python / PyTorch) for large-scale training on two of the world's most powerful supercomputers (InstaDeep's Kyber, ~500 PFLOPs, and EuroHPC's MareNostrum 5, 260 PFLOPs peak), as well as GPUs and the cloud, with training and data pipelines orchestrated in Dagster.

Barco, Senior Software Development Engineer, Machine Learning

Kortrijk, Belgium

Feb 2020 – Aug 2022

2 years 7 months

- Built a TensorFlow Extended production pipeline (orchestrated with Apache Airflow) training deep-learning models on multispectral images to detect and classify melanoma skin cancers for Demetra, Barco's dermatology imaging device.
- Engineered the pipeline with end-to-end data and model lineage tracking to deliver the traceability and reproducibility that regulated medical industries (FDA approval) require.
- Worked across the research, cloud-backend and mobile/web frontend teams.

Bayer Crop Science, then BASF, Bioinformatics Researcher, Manager of the Python & R Platforms

Zwijnaarde, Belgium

Feb 2018 – Jan 2020

2 years

- Owned the Python/R data-analysis platform used by 480 researchers and data scientists: technical support, proactive monitoring and root-cause analysis, training (NumPy, pandas, BioPython) and mentoring across the company's internal research community.
- The Zwijnaarde site was divested from Bayer Crop Science to BASF as part of Bayer's acquisition of Monsanto; I continued in the same role across the transition.

Applied Maths (acquired by bioMérieux), Bioinformatics Software Developer

Sint-Martens-Latem, Belgium

Sept 2011 – Jan 2018

6 years 5 months

- Worked on BioNumerics, a bioinformatics suite for integrated analysis of biological data, writing custom Python scripts and extensions for clients in the clinical, academic and government sectors.

Selected publications

InstaNovo enables diffusion-powered *de novo* peptide sequencing in large-scale proteomics experiments

2025

Trained on ~63 million spectra; top-ranked in an independent benchmark of 17 *de novo* tools across 83 datasets; Altmetric attention score 100+ and 120+ GitHub stars.

K. Eloff, K. Kalogeropoulos, ..., **J. Van Goey**, ..., T. P. Jenkins

[10.1038/s42256-025-01019-5](https://doi.org/10.1038/s42256-025-01019-5) (Nature Machine Intelligence)

InstaNovo-P: a *de novo* peptide sequencing model for phosphoproteomics

2025

J. Lauridsen, P. Ramasamy, ..., **J. Van Goey**, ..., K. Kalogeropoulos

[10.1101/2025.05.14.654049](https://doi.org/10.1101/2025.05.14.654049) (bioRxiv)

De novo peptide sequencing rescoring and FDR estimation with Winnow	2025
A. Mabona, J. Daniel, ..., J. Van Goey , K. Kalogeropoulos 10.48550/arXiv.2509.24952 (arXiv)	
A living proteomics benchmark for comprehensive evaluation of deep learning-based de novo peptide sequencing tools	2026
M. Pominova, J. Van Goey , C. Adams, ..., W. Bittremieux 10.6084/m9.figshare.31680883.v1 (Registered report (preprint))	
Generalizable direct protein sequencing with InstaNexus	2026
M. Reverenna, M. Wennekers Nielsen, ..., J. Van Goey , ..., K. Kalogeropoulos 10.1016/j.mcpro.2026.101547 (Molecular & Cellular Proteomics)	
A Framework for Database Search with AI Models in Mass Spectrometry-Based Proteomics	2026
K. Kalogeropoulos, J. Van Goey , T. P. Jenkins, K. M. Eloff 10.1021/acs.jproteome.5c01153 (Journal of Proteome Research)	

Education

Microdegree	Hogeschool West-Vlaanderen (Howest) , Machine Learning & Deep Learning	Belgium 2018 – 2019
	Katholieke Universiteit Leuven , Bioinformatics Partial credits obtained.	Leuven, Belgium 2010 – 2012
M.Sc.	University of Antwerp , Biology	Antwerp, Belgium 1997 – 2004

Skills

Machine learning: Transformers, diffusion models, large language models (LLMs), supervised & self-supervised learning
Frameworks: PyTorch, TensorFlow / TFX, NumPy, pandas
Scale & MLOps: Distributed GPU training, HPC (InstaDeep's Kyber and EuroHPC's MareNostrum 5 supercomputers), Kubernetes-based ML platforms (Alichor), cloud (AWS, GCP), Docker, CI/CD, workflow orchestration (Airflow, Dagster), experiment tracking (MLflow, Neptune, TensorBoard)
Bioinformatics: BioPython, BLAST, ClustalW, Snakemake, BioNumerics
Languages: Python (primary), R

Projects

Awesome De Novo Peptide Sequencing

A comprehensive, interactive map of the *de novo* peptide sequencing field: algorithms, post-processors, downstream applications, and adjacent tools, deep-learning and classical alike.

- Open source: github.com/BioGeek/awesome_de_novo_peptide_sequencing

Teaching & mentoring

Machine Learning for Biology practical	Deep Learning Indaba, Ghana 2023
Mentored the hands-on practical alongside InstaDeep and Google DeepMind colleagues.	
Scientific Python training & mentoring	BASF and Bayer Crop Science, Belgium 2018 – 2020
Trained and mentored researchers and data scientists in scientific Python (NumPy, pandas, BioPython) as manager of the Python/R analysis platform.	

Hackathons (co-created, organized and judged)

2025: Machine Learning Interatomic Potentials Hackathon, IndabaX (Stellenbosch University, South Africa)
2024: Snakes and Sequences: Senegalese Serpent Venom Sequencing Hackathon, Deep Learning Indaba (Senegal)
2024: Desert Locust Breeding Ground Prediction Hackathon, IndabaX (Wits, Johannesburg)
2023: Unveiling Cassava's Secrets, Deep Learning Indaba (Ghana)